

✓ Customer Segmentation Using Machine Learning

Algonive Data Analytics Internship Project

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Domain: Data Analytics

Goal: Segment customers based on income and spending behavior to help businesses target marketing strategies.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
!ls
```

```
archive.zip  Mall_Customers.csv  sample_data
```

```
!unzip archive.zip
```

```
Archive:  archive.zip
  inflating: Mall_Customers.csv
```

```
df = pd.read_csv('Mall_Customers.csv')
df.head()
```

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)	
0	1	Male	19	15	39	
1	2	Male	21	15	81	
2	3	Female	20	16	6	
3	4	Female	23	16	77	
4	5	Female	31	17	40	

Next steps:

[Generate code with df](#)[New interactive sheet](#)

Dataset contains the following columns:

CustomerID – Unique customer number

Genre – Male/Female

Age – Customer age

Annual Income – Income per year

Spending Score – Score given based on customer spending behavior

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   CustomerID                            200 non-null    int64
1   Genre                                 200 non-null    object
2   Age                                   200 non-null    int64
3   Annual Income (k$)                    200 non-null    int64
4   Spending Score (1-100)                 200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

```
df.describe()
```

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

```
df.isnull().sum()
```

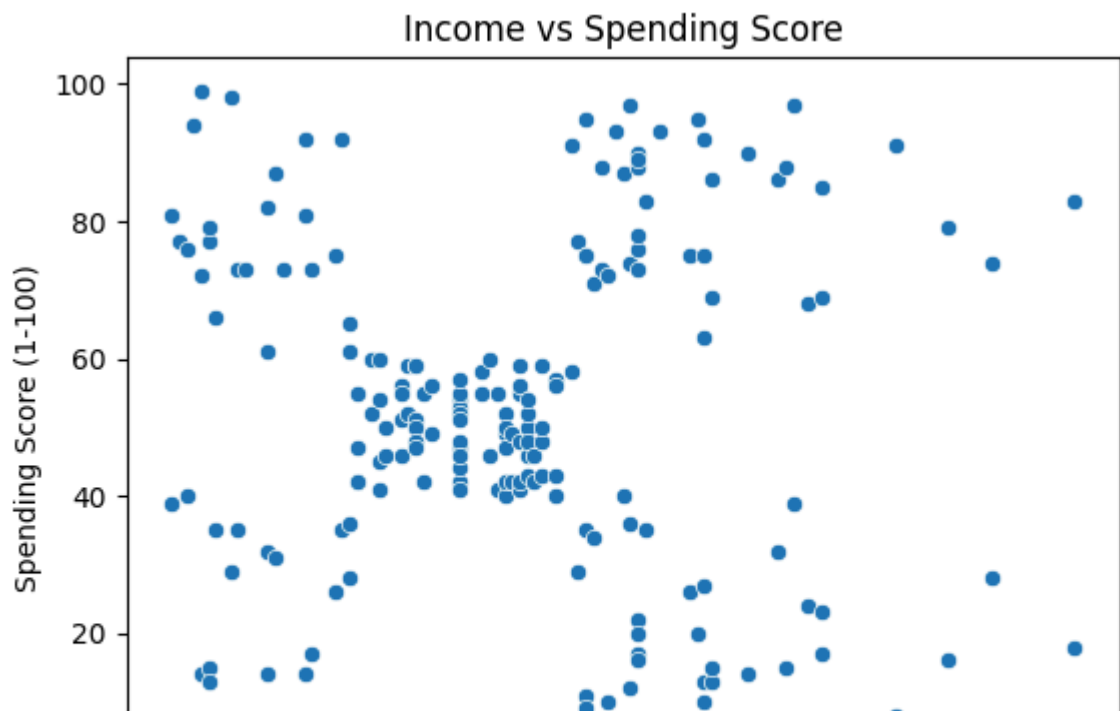
	0
CustomerID	0
Genre	200
Age	0
Annual Income (k\$)	0
Spending Score (1-100)	0
Cluster	0

```
df['Genre'] = df['Genre'].map({'Male':0,'Female':1})
```

EDA helps understand customer behavior patterns.

We analyze income vs spending score relationship.

```
sns.scatterplot(x='Annual Income (k$)', y='Spending Score (1-100)', data=df)
plt.title("Income vs Spending Score")
plt.show()
```

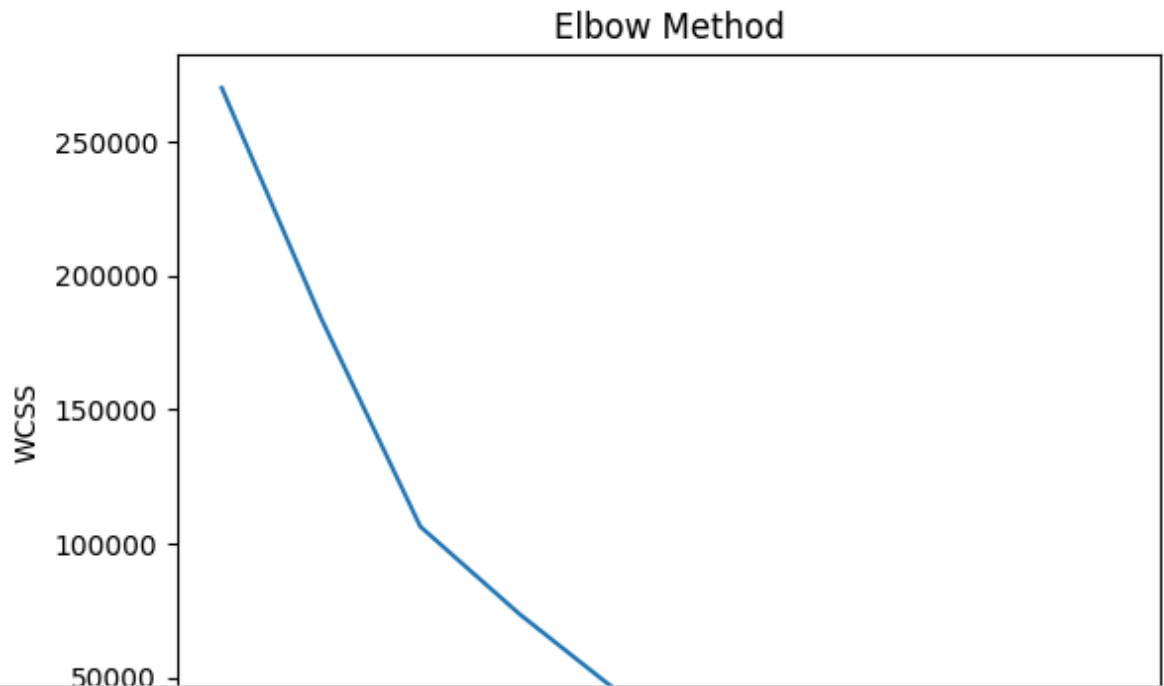


```
from sklearn.cluster import KMeans

wcss = []
for i in range(1,11):
    kmeans = KMeans(n_clusters=i)
    kmeans.fit(df[['Annual Income (k$)', 'Spending Score (1-100)']])
    wcss.append(kmeans.inertia_)

plt.plot(range(1,11), wcss)
```

```
plt.title("Elbow Method")
plt.xlabel("Number of clusters")
plt.ylabel("WCSS")
plt.show()
```

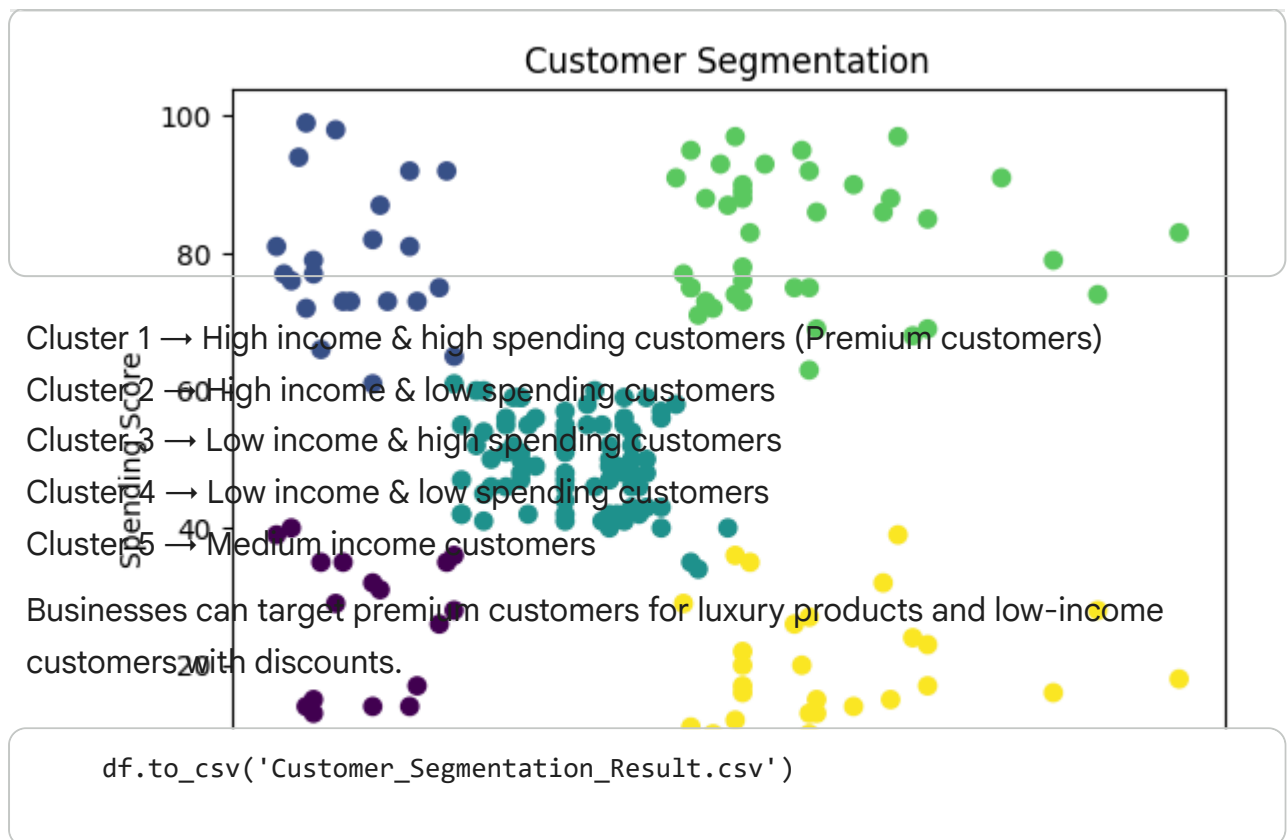


K-Means clustering algorithm is used to group customers into segments based on similar income and spending patterns.

```
X = df[['Annual Income (k$)', 'Spending Score (1-100)']]
```

```
kmeans = KMeans(n_clusters=5)
df['Cluster'] = kmeans.fit_predict(X)
```

```
plt.scatter(df['Annual Income (k$)'], df['Spending Score (1-100)'], c=df['Cluster'])
plt.xlabel("Income")
plt.ylabel("Spending Score")
plt.title("Customer Segmentation")
plt.show()
```



This project successfully segments customers based on their purchasing behavior using Machine Learning.

The results help businesses understand customer categories and make better marketing decisions.